YASHVI VAGHELA

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EDUCATION

University of Southern California, Los Angeles **August 2023 - May 2025**
Master of Science in Computer Science **(GPA 3.81/4)**

University of Mumbai, Mumbai **August 2019 - May 2023**
Bachelor of Engineering in Computer Engineering **(GPA 9.35/10)**

EXPERIENCE

Easley-Dunn Productions Inc- Software Developer **July 2025 - May 2026**

- Built a full-stack video annotation platform in React, enabling consistent labeling and CSV export of structured gameplay datasets, cutting manual labeling time by 50% through frame-accurate scrubbing and hotkey-based tagging.
- Enabled the AI team to tune model parameters using structured move-probability and bounding-box data, by integrating computer vision analysis into a Flask backend and processing 50+ gameplay videos.
- Eliminated redundant video reprocessing by designing a caching architecture that reused prior analysis results, reducing repeat-view compute time to near-zero and powering an interactive dashboard with live heatmaps and probability charts.
- Led a 5-person cross-functional engineering team, aligning frontend and backend workstreams and maintaining code documentation to streamline onboarding.

Ikigai-The Art of Living, India- *UX/UI Intern* **June 2022 - July 2022**

- Designed and prototyped 5 responsive web interfaces and reusable UI components in Figma, including navigation flows, input controls, and layouts optimized for desktop and mobile experiences.
- Collaborated with developers to translate designs into production-ready interfaces, improving component consistency, streamlining developer handoff, and contributing to a 20% reduction in front-end bugs.

PROJECTS

StudyBot *(Python, FastAPI, LangChain, Groq API, Qdrant, SQLite, Next.js, Gemini)* | [GitHub]

- Built a full-stack RAG-based study assistant using LangChain and Groq API to generate AI-based study plans and stream real-time chat responses, adapting explanations based on struggle patterns detected within each topic.
- Designed a FastAPI backend with SQLite and a vector DB for sub-500ms semantic retrieval, referencing struggling topics, detecting prerequisite gaps, and surfacing related concepts, with plans adjustable via natural conversation.

Multi-Agent Resume Tailoring System (Python, LLMs, Agentic AI, Streamlit)| [GitHub]

- Built a multi-agent resume-tailoring system with an executor agent to draft content, a grounding critic to independently verify each bullet against real project data, and a Fit Scorer to evaluate fit and trigger revisions.
- Demonstrated a 65.5-percentage-point reduction in fabricated resume claims by testing the pipeline with and without a grounding-critic step on identical test cases.

Stock Search Website *(MongoDB, Express, Angular, Node.js, GCP)* | [GitHub]

- Developed a stock search web application using Angular and Node.js, integrating Finnhub and Polygon APIs to display real-time market data for 50+ stocks.
- Implemented portfolio simulation features including buy/sell transactions and watchlists, with user data stored in MongoDB, backend services deployed on Google App Engine, and caching added for frequently requested ticker data to reduce redundant API calls and improve response consistency.

Interview Feedback Assistant *(Flask, Python, NLP, spaCy)* | [GitHub]

- Built an interview feedback platform that analyzed recorded responses and generated automated feedback using NLP pipelines.
- Implemented resume-job description similarity scoring with TF-IDF and cosine similarity, and used spaCy to detect speech patterns and undesirable phrases in interview responses.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, Java, JavaScript, TypeScript, SQL
- **Frameworks & Web:** React.js, Angular, Next.js, Node.js, Express.js, Django, Flask, FastAPI, REST APIs, Streamlit
- **Databases:** PostgreSQL, MySQL, MongoDB, SQLite, Oracle, Qdrant
- **AI/ML:** LLMs, RAG, Gemini, Groq API, LangChain, TensorFlow, OpenCV, NumPy, Vector Databases
- **Tools & Platforms:** AWS, Docker, Git, GCP (Google App Engine), Figma, Pydantic
- **Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Design, Natural Language Processing (NLP), Cloud Computing, System Design, CI/CD, Agentic AI Systems

RESEARCH PAPERS

Implementation of Applied Machine Learning for Identification of Exoplanets Using NASA's API

- Developed and evaluated ML models on NASA's Kepler dataset for exoplanet classification, achieving 91% accuracy with Random Forest after SMOTE-based class balancing; findings were published in the International Journal of Design Engineering (2021).